

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 3.2: Rotation — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.2: Rotation
Driving Question	DRIVING QUESTION: What do you think makes it possible for a daily pattern of day and night experience on Earth?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Phenomenon: Day and Night over Kenya	We watched a time-lapse of Earth where 24 hours were shown in 24 seconds (1 second = 1 hour). We saw dawn over Mombasa at about 6:30 am, midday over Nairobi at about 12:30 pm, sunset over Lake Victoria at about 6:40 pm and darkness over Kisumu at about 7:30 pm, and the pattern repeated every 24 hours. With a globe and torch, we saw that turning the globe makes places move from light into dark.	A turning can be described using a centre, an angle and a direction. A full turn is 360° , a half turn is 180° and a quarter turn is 90° . Positive angles are anticlockwise and negative angles are clockwise. Kenya feels still, but the video suggests a steady turning, and Mombasa meets the light before Kisumu.	Our first explanation is that Earth turns about its axis so that some parts face the Sun and have day while the other side is dark and has night. We still need to find out why Mombasa gets sunrise before Kisumu and how to describe the turning with a centre, an angle and a direction. Our questions are on the Driving Question Board.
Lesson 2 Properties of Rotation: What Stays the Same and What Turns?	We rotated triangle ABC with A(2,1), B(4,1), C(4,3) through 90° anticlockwise about O(0,0) on tracing paper and on the Cartesian plane, and we obtained A'(-1,2), B'(-1,4), C'(-3,4). We measured $OA = OA' = \sqrt{5}$, $OB = OB' = \sqrt{17}$, $OC = OC' = 5$, and sides $AB = A'B' = 2$, $BC = B'C' = 2$, $AC = A'C' = \sqrt{8}$. A turn of -90° (clockwise) sent A to A''(1,-2).	A rotation turns every point about a fixed centre through the same angle in the same direction. Distances from the centre, lengths, angles, shape and size stay the same, so the image is congruent to the object. Only the position and the direction the figure faces change. Positive angles are anticlockwise and negative angles are clockwise.	Kenya feels still but turns with Earth about its axis. Because rotation preserves distances, shape and size, Nairobi, Mombasa and Kisumu keep their distances from the axis and from one another as they turn anticlockwise together, which helps explain why the day-night pattern repeats regularly. The turning is described with a centre (Earth's axis), an angle and a direction.
Lesson 3 Finding the Centre and Angle of Rotation from an Object and Its Image	We constructed the perpendicular bisectors of PP', QQ' and RR' on a grid. The bisectors of PP' and RR' met at O(0,0), and the bisector of QQ' lay on the same line as that of PP' ($y = x$). Tracing paper confirmed that	The centre of rotation lies where the perpendicular bisectors of object-image pairs meet, and at least two different bisectors are needed. The angle is measured from the centre to the object	Earth's rotation can be described by a centre (its axis), an angle (15° per hour) and a direction (anticlockwise as seen from above the North Pole). Kenya is not still: it turns with Earth, and that steady turning

	turning through 90° anticlockwise about O takes P to P'. In the video, 6:30 am Mombasa dawn to 12:30 pm Nairobi midday is 6 hours.	point and then to the image point, positive if anticlockwise. Earth turns 15° per hour, so 90° takes 6 hours.	brings Mombasa, Nairobi and Kisumu in turn into the Sun's light, creating day and night every 24 hours.
Lesson 4 Rotations on Grids: Positive and Negative Angles, Clockwise and Anticlockwise Turns	We constructed images of triangle ABC about O(0,0) using tracing paper and grid paper: 90° anticlockwise gave A'(-1,2), B'(-1,4), C'(-3,4); -90° (90° clockwise) gave A''(1,-2); a half turn changed (x,y) to (-x,-y). Distances from O and the sides of the triangle stayed the same.	A positive angle means an anticlockwise turn and a negative angle means a clockwise turn. Rules about O: 90° anticlockwise (x,y) → (-y,x); -90° (x,y) → (y,-x); 180° (x,y) → (-x,-y). Any rotation needs a centre, an angle and a direction.	Earth's turning is a rotation with centre at its axis, angle 15° per hour and direction anticlockwise (positive) as seen from above the North Pole. In 6 hours it turns 90° and in 12 hours 180°. Kenya turns with Earth, and the Sun only appears to move from east to west, which helps explain the regular day and night pattern.
Lesson 5 Order of Rotational Symmetry of Plane Figures: How Many Times Does a Figure Fit Onto Itself in One Full Turn?	We turned a square, equilateral triangle, regular hexagon, rectangle, regular pentagon, isosceles triangle, a kiondo pattern, an 8-spoke bicycle wheel and a 5-nut matatu wheel through one full turn of 360° and counted how many times each fitted onto itself. Orders found: square 4, triangle 3, hexagon 6, rectangle 2, pentagon 5, isosceles triangle 1, kiondo 6, bicycle wheel 8, matatu wheel 5.	The order of rotational symmetry is the number of times a figure fits onto itself in one full turn of 360°. The smallest angle of fit = $360^\circ \div \text{order}$ (e.g. square 90°, hexagon 60°, 8-spoke wheel 45°, matatu wheel 72°). Number of sides does not always equal the order (a rectangle has order 2). A figure with order 1 has no rotational symmetry.	A full turn of 360° can be divided into equal steps. In the same way Earth turns 360° in 24 hours, which is $360^\circ \div 24 = 15^\circ$ per hour, anticlockwise about its axis. This regular turning is part of why Kenya, from Mombasa to Nairobi to Kisumu, has a repeating daily pattern of day and night, even though Kenya feels still.
Lesson 6 Axis and Order of Rotational Symmetry of Solids: From Cuboids to Earth's Axis	What we observed: a cuboid with non-square ends turned onto itself twice in a full turn about each of 3 axes; a square-based prism had one axis of order 4 and two of order 2; a cone fitted at every angle about its axis; the globe turned about its axis through the North and South Poles, anticlockwise from above the North Pole, with Mombasa meeting the light before Kisumu.	What we learned: an axis of rotational symmetry is a line about which a solid fits onto itself, its order is the number of fits in 360°, and the smallest angle of fit is $360^\circ \div \text{order}$ (cuboid 3 axes of order 2; square prism order 4 and order 2; cone and sphere order infinity). Earth's axis through the poles is the centre of its rotation at 15° per hour.	How this helps explain the phenomenon: Day and night repeat every 24 hours because Earth rotates once (360°) about its axis, turning anticlockwise at 15° per hour while the Sun lights one half. The axis is the centre, 15° per hour is the angle and anticlockwise is the direction; Kenya is not still but turning with Earth, so Mombasa reaches sunrise before Kisumu.
Lesson 7 Congruence and Rotation: Is the Image Really the Same Shape? (Why Kenya Can Turn Without Changing)	We observed that triangle ABC with A(2,1), B(4,1), C(4,3) and its 90 degree anticlockwise image A'(-1,2), B'(-1,4), C'(-3,4) have equal sides (2, 2, square root of 8) and equal angles, and the traced triangle fits exactly after turning.	We learned that rotation preserves lengths, angles, size and shape, so the object and image are directly congruent. Only the position and direction of facing change.	We can explain that Kenya and its cities turn with Earth at 15 degrees per hour anticlockwise without being stretched or squashed, because rotation is a direct isometry, so Nairobi, Mombasa and Kisumu keep their size and shape while day and night sweep across them.
Lesson 8 Applications in Real-Life Situations: Wheels, Turbines, Clocks and	We observed that the minute hand turns 6° per minute and the hour hand 30° per hour (clockwise, so negative angles), the Lake Turkana turbine has order 3 (120°), Mzee	We learned that any turning can be described by a centre, an angle and a direction, that the smallest angle of fit is $360^\circ \div \text{order}$, and that rate × time gives the	We can explain that the day and night pattern over Mombasa, Nairobi and Kisumu fits the same kind of turning as a clock or wheel: Earth rotates 360° in 24 h about its

Kiondo Patterns	Kamau's 8-spoke wheel has order 8 (45°), and Mama Achieng's kiondo pattern has order 6 (60°).	angle turned, just as Earth turns 15° per hour anticlockwise.	axis at 15° per hour, so places further east like Mombasa turn into sunlight sooner than places further west like Kisumu.
Lesson 9 Revisiting the Phenomenon: Explaining Day and Night over Mombasa, Nairobi and Kisumu with Rotation	We observed that when the globe is turned anticlockwise 15° for each hour with the torch fixed, the Mombasa dot enters the light first, then Nairobi, then Kisumu. The time-lapse also showed dawn over Mombasa, midday over Nairobi, sunset over Lake Victoria and darkness over Kisumu.	We learned that Earth rotates 360° in 24 hours, which is 15° per hour, anticlockwise when viewed from above the North Pole, about a centre that is Earth's axis. The longitude differences of 2.9° , 2.0° and 4.9° give time differences of about 12, 8 and 20 minutes.	Day and night repeat every 24 hours because Earth rotates once about its axis in 24 hours while the Sun lights one half. Mombasa is further east than Kisumu, so it turns into the sunlight about 20 minutes sooner. The Sun only appears to move east to west; Kenya is turning with Earth.
Lesson 10 Consolidation and Performance Task: Designing a Rotationally Symmetric Pattern and the Final Explanation of Day and Night	What we observed: In the time-lapse, dawn came to Mombasa (about 6:30 am) first, then midday over Nairobi (about 12:30 pm), sunset over Lake Victoria (about 6:40 pm) and darkness over Kisumu (about 7:30 pm), and the pattern repeated every 24 hours. We also observed that our patterns fit onto themselves several times in one full turn, and that the globe with a torch lit one half at all times as it turned.	What we learned: A rotation is described by a centre, an angle and a direction (positive = anticlockwise, negative = clockwise). The order of rotational symmetry is the number of times a figure fits onto itself in 360° , and the smallest angle is $360^\circ \div \text{order}$ (e.g. kiondo order 6 at 60° , 8-spoke wheel order 8 at 45° , Lake Turkana turbine order 3 at 120°). Earth turns 360° in 24 hours, which is 15° per hour.	How we can explain it: Day and night repeat every 24 hours because Earth rotates once (360°) about its axis in 24 hours, turning anticlockwise at 15° per hour as seen from above the North Pole, while the Sun always lights one half. Mombasa is further east than Kisumu (4.9° difference, $4.9 \div 15$ is about 20 minutes), so it turns into sunlight about 20 minutes sooner. The Sun only appears to move east to west, and Kenya is not still because it turns with Earth.

ATTRIBUTION AND LICENSING

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